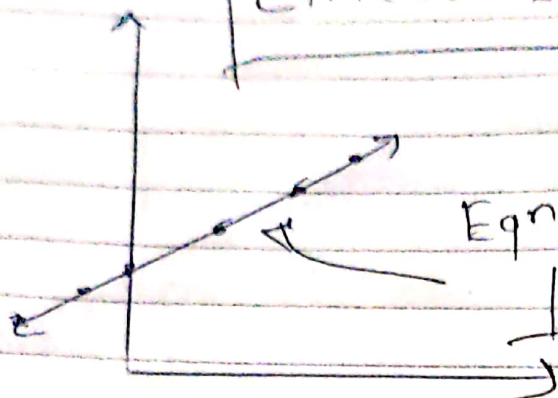


Linear Equation



Egn of line is

$$y = mx + c$$

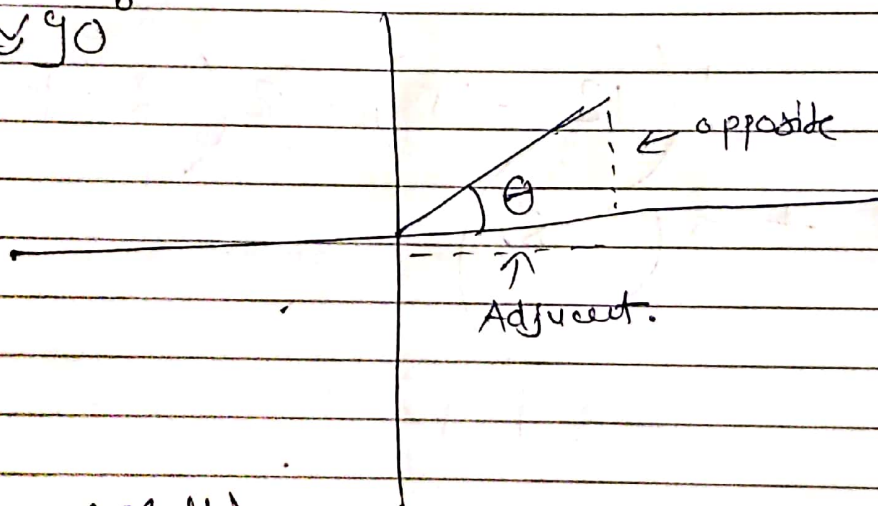
$$y = mx + c$$

interception [constant]

c add in every step as constant.

is when c value is a y value when $x=0$

$$\tan^{-1}(\infty) \approx 90^\circ$$



m is
Slope

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

- ① θ is +ve in 1st quadrant
- ② In 2nd quadrant x is -ve so $\tan^{-1}\left(\frac{y}{-x}\right)$ in 90° to 180°
- ③ in 3rd quadrant $\tan^{-1}\left(\frac{-y}{-x}\right)$ +ve in $(180^\circ - 270^\circ)$
- ④ In 4th quadrant $\tan^{-1}\left(\frac{-y}{x}\right)$ -ve in $(270^\circ$ to $360^\circ)$

Teacher's Signature: